# Intro to Controversy Mapping

## 42620 Science, Technology & Society

**Anders Kristian Munk** | Human-Centered Innovation | Department of Technology, Management and Economics

Controversies constantly produce **new stakeholders** who were not previously aware of their interests and did not express their views on the problem. In other words: **not on our radar**.

# This is a challenge for RRI…

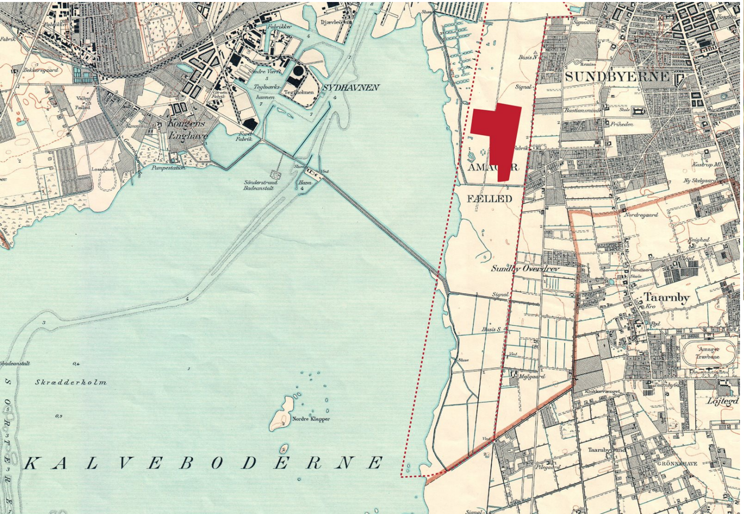
- We are supposed to open up anticipation and reflection to a broad variety of relevant actors (engage). **But who are these actors?**
- We know that problem wickedness depends on factors like how actors frame the problems. **So how are they framing them?**

**Controversy mapping answers some simple and descriptive questions in a situation where the ground is shifting under our feet:**

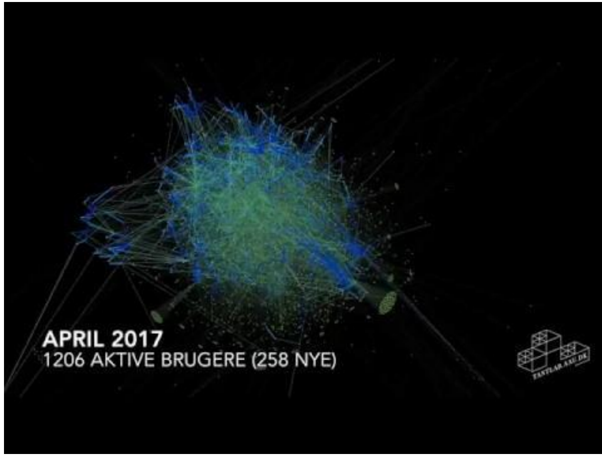
- Who are the actors?
- What are their concerns?
- What kind of arguments are they making?
- And how is it changing?

Using quali-quantitative methods to visualize and explore - data-driven and from the ground up.

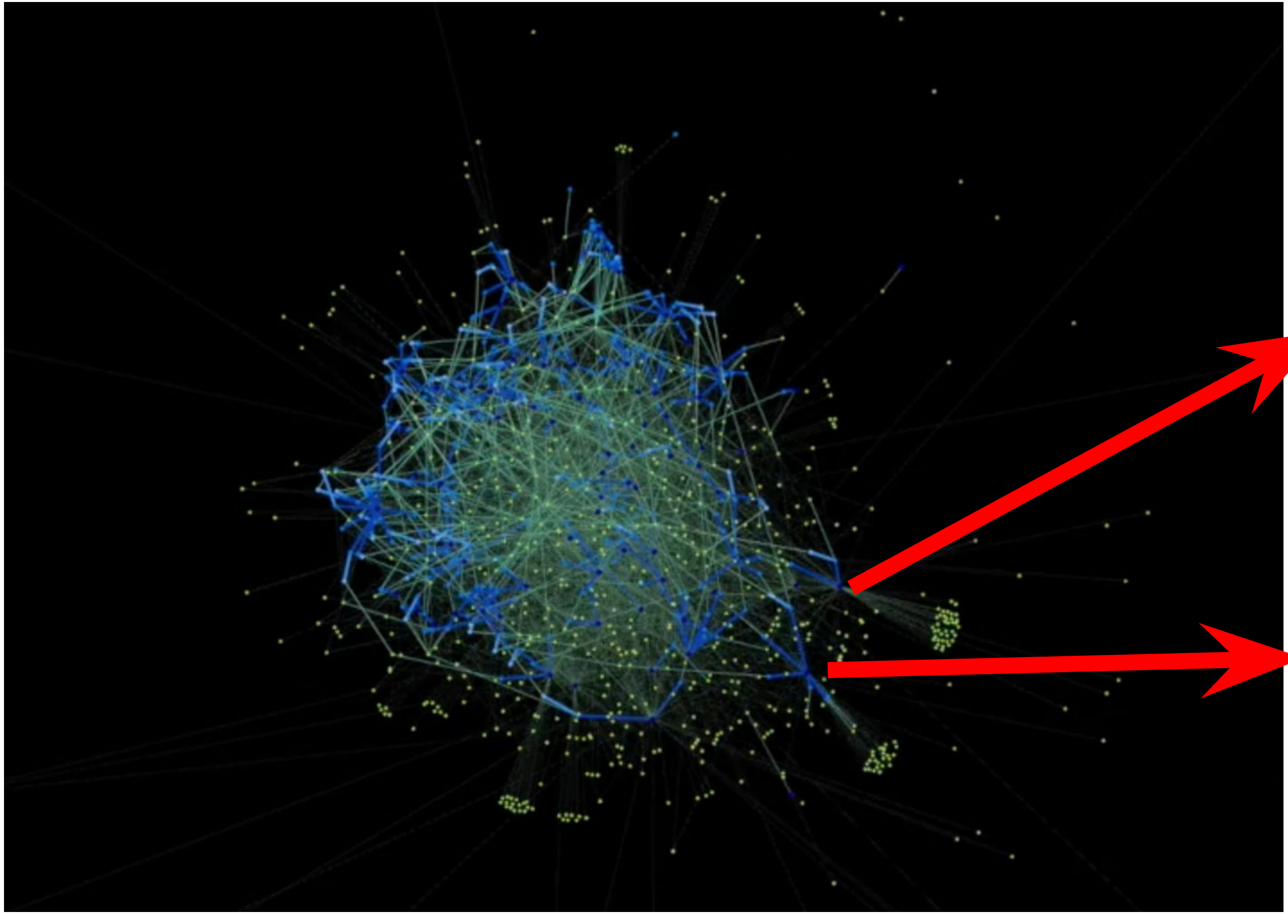
Examples of quali-quantitative exploration of controversies to discover actors and issues as they emerge.








The battle for Amager Commons (Fælled), 2016/2017



Seneste: Torsdag nu oppe på over 28.000 underskrifter 🙂 Snart sidste frist. HUSK at SKRIVE UNDER og DEL meget gerne. Vi har kun indtil den 13. oktober hvor underskrifterne vil blive overleveret til Borgerrepræsentationen mfl. Tak 🙂

http://www.skrivunder.net/bevar_koebenhavns_naturhistoriske...


Like Comment Share

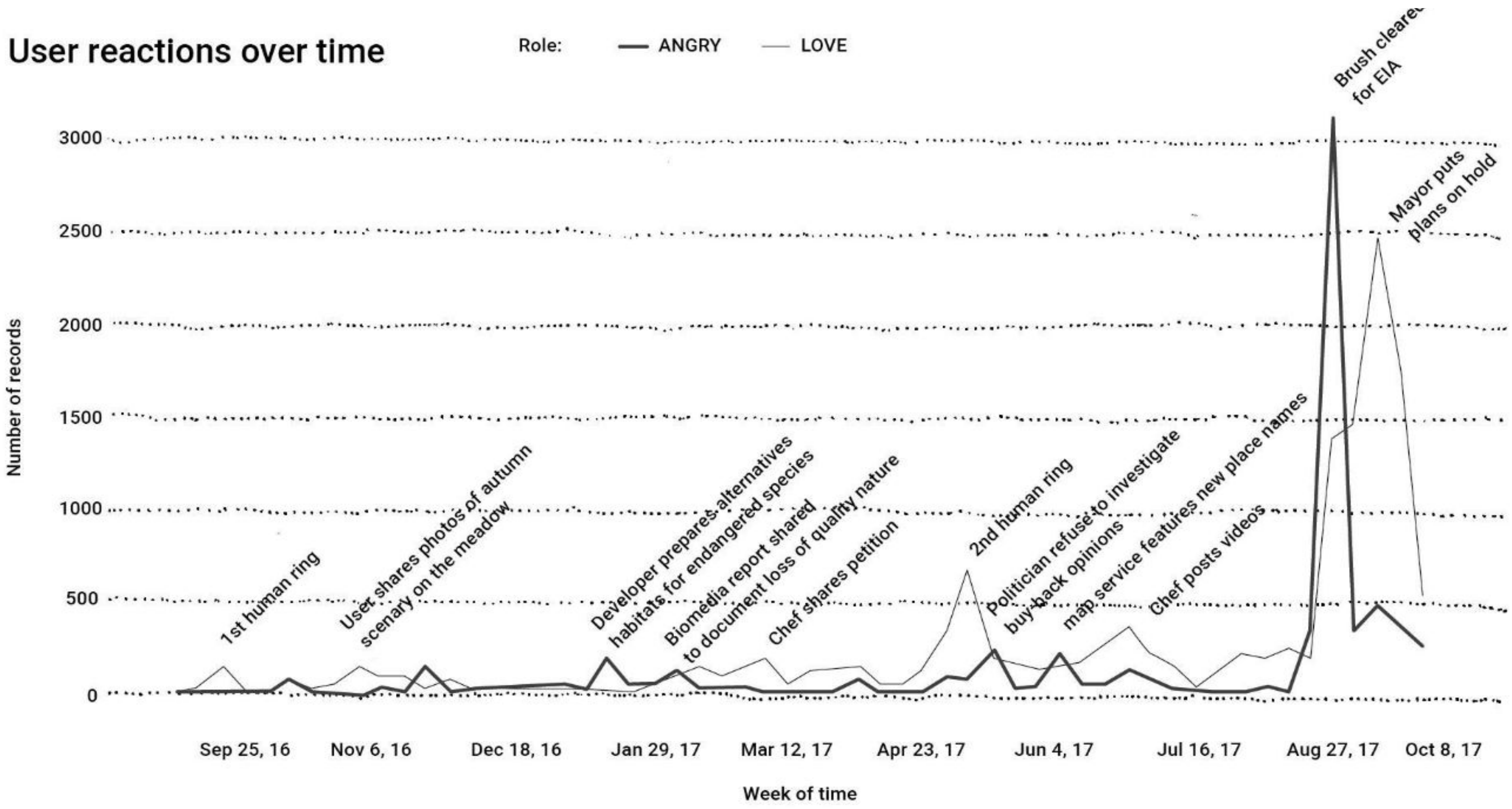
238

HUSK at SKRIVE UNDER og DEL meget gerne. Vi har kun indtil den 13. oktober hvor underskrifterne vil blive overleveret til Borgerrepræsentationen. Tak 🙂

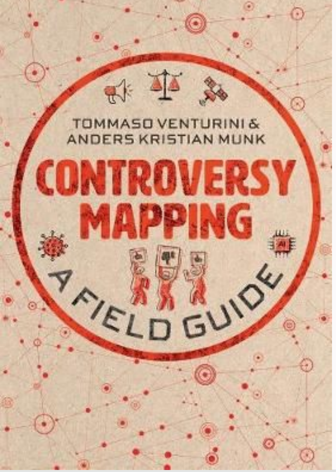
http://www.skrivunder.net/bevar_koebenhavns_naturhistoriske...

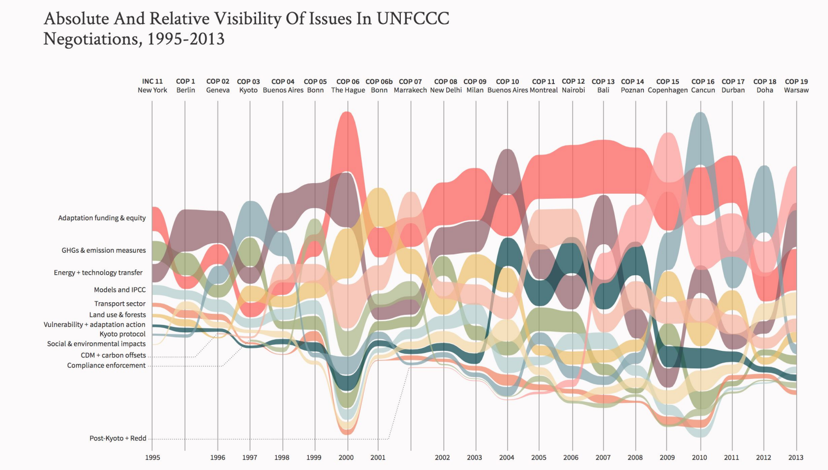

Like Comment Share

269



User reactions over time
Role:
ANGRY
LOVE
Number of records
3000
2500
2000
1500
1000
500
0
1st human ring
User shares photos of autumn
scenary on the meadow
Developer prepares alternatives
habitats for endangered species
Biomedia report shared
to document loss of quality nature
Chef shares petition
2nd human ring
Politician refuse to investigate
buy-back opinions
map service features new place names
Chef posts videos
Brush cleared
for EIA
Mayor puts
plans on hold
Sep 25, 16
Nov 6, 16
Dec 18, 16
Jan 29, 17
Mar 12, 17
Apr 23, 17
Jun 4, 17
Jul 16, 17
Aug 27, 17
Oct 8, 17
Week of time


TOMMASO VENTURINI &
ANDERS KRISTIAN MUNK
CONTROVERSY
MAPPING
A FIELD GUIDE


Absolute And Relative Visibility Of Issues In UNFCCC Negotiations, 1995-2013
INC 11 New York
COP 1 Berlin
COP 02 Geneva
COP 03 Kyoto
COP 04 Buenos Aires
COP 05 Bonn
COP 06 The Hague
COP 06b Bonn
COP 07 Marrakech
COP 08 New Delhi
COP 09 Milan
COP 10 Buenos Aires
COP 11 Montreal
COP 12 Nairobi
COP 13 Bali
COP 14 Poznan
COP 15 Copenhagen
COP 16 Cancun
COP 17 Durban
COP 18 Doha
COP 19 Warsaw
Adaptation funding & equity
GHGs & emission measures
Energy + technology transfer
Models and IPCC
Transport sector
Land use & forests
Vulnerability + adaptation action
Kyoto protocol
Social & environmental impacts
CDM + carbon offsets
Compliance enforcement
Post-Kyoto + Redd
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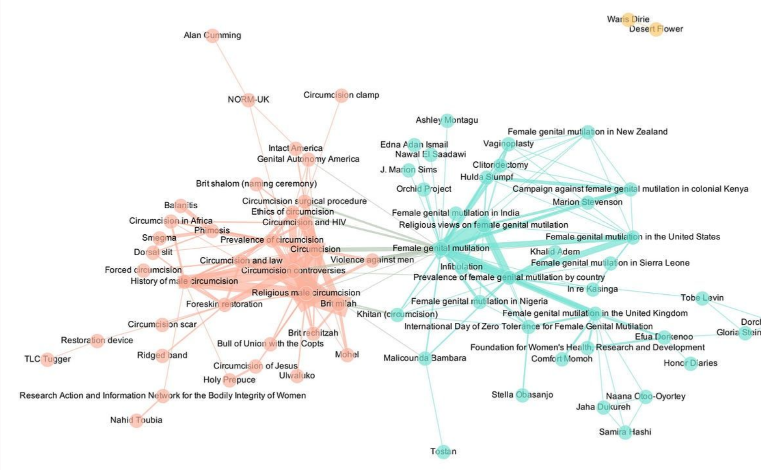
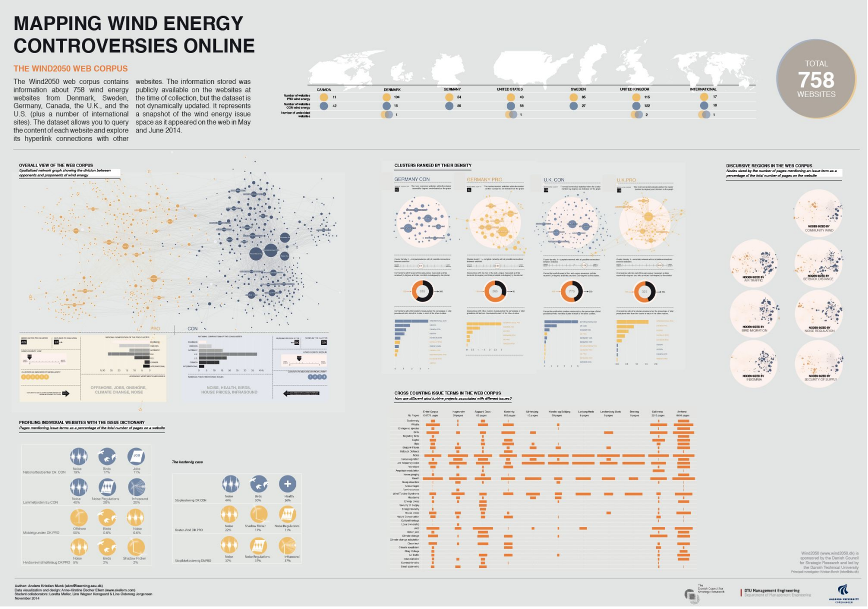
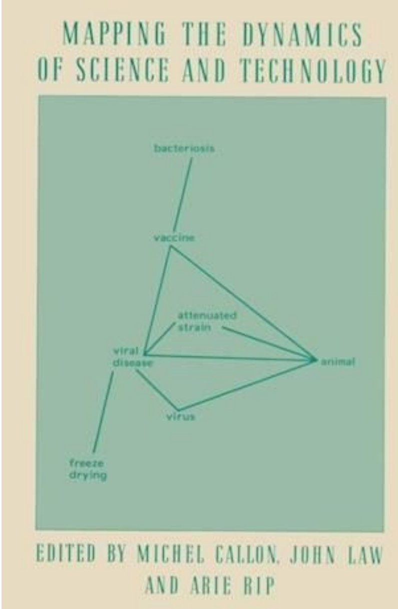
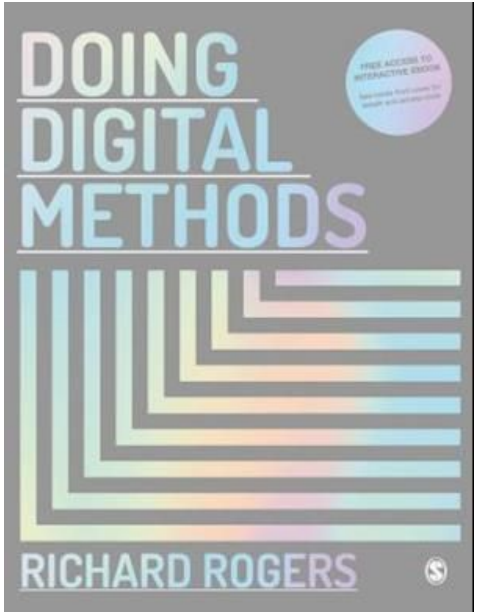

Alan Cumming
Waris Dirie
Desert Flower
NORM-UK
Circumcision clamp
Ashley Montagu
Female genital mutilation in New Zealand
Intact America
Genital Autonomy America
Edna Adan Ismail
Nawal El Saadawi
Vaginoplasty
Clitoridectomy
J. Marion Sims
Hulda Stumpf
Brit shalom (naming ceremony)
Orchid Project
Campaign against female genital mutilation in colonial Kenya
Balanitis
Circumcision surgical procedure
Marion Stevenson
Ethics of circumcision
Female genital mutilation in India
Circumcision in Africa
Circumcision and HIV
Religious views on female genital mutilation
Phimosis
Smegma
Prevalence of circumcision
Female genital mutilation in the United States
Dorsal slit
Circumcision
Female genital mutilation
Khalid Adem
Circumcision and law
Violence against men
Female genital mutilation in Sierra Leone
Forced circumcision
Circumcision controversies
Infibulation
History of male circumcision
Prevalence of female genital mutilation by country
In re Kasinga
Religious male circumcision
Tobe Levin
Foreskin restoration
Brit milah
Female genital mutilation in Nigeria
Female genital mutilation in the United Kingdom
Khitan (circumcision)
Circumcision scar
International Day of Zero Tolerance for Female Genital Mutilation
Brit rechitzah
Efua Dorkenoo
Restoration device
Bull of Union with the Copts
Foundation for Women's Health, Research and Development
TLC Tugger
Ridged band
Mohel
Malicounda Bambara
Comfort Momoh
Honor Diaries
Circumcision of Jesus
Holy Prepuce
Ulwaluko
Research Action and Information Network for the Bodily Integrity of Women
Stella Obasanjo
Naana Otoo-Oyortey
Jaha Dukureh
Nahid Toubia
Samira Hashi
Tostan


MAPPING WIND ENERGY CONTROVERSIES ONLINE
THE WIND2050 WEB CORPUS
The Wind2050 web corpus contains information about 758 wind energy websites from Denmark, Sweden, Germany, Canada, the U.K., and the U.S. (plus a number of international sites). The dataset allows you to query the content of each website and explore its hyperlink connections with other websites. The information stored was publicly available on the websites at the time of collection, but the dataset is not dynamically updated. It represents a snapshot of the wind energy issue space as it appeared on the web in May and June 2014.
TOTAL
758
WEBSITES
OVERALL VIEW OF THE WEB CORPUS
CLUSTERS RANKED BY THEIR DENSITY
DISCURSIVE REGIONS IN THE WEB CORPUS
CROSS COUNTING ISSUE TERMS IN THE WEB CORPUS
PROFILING INDIVIDUAL WEBSITES WITH THE ISSUE DICTIONARY


MAPPING THE DYNAMICS
OF SCIENCE AND TECHNOLOGY
bacteriosis
vaccine
attenuated strain
viral disease
animal
virus
freeze drying
EDITED BY MICHEL CALLON, JOHN LAW
AND ARIE RIP


DOING
DIGITAL
METHODS
RICHARD ROGERS

# MAPPING WIND ENERGY CONTROVERSIES ONLINE

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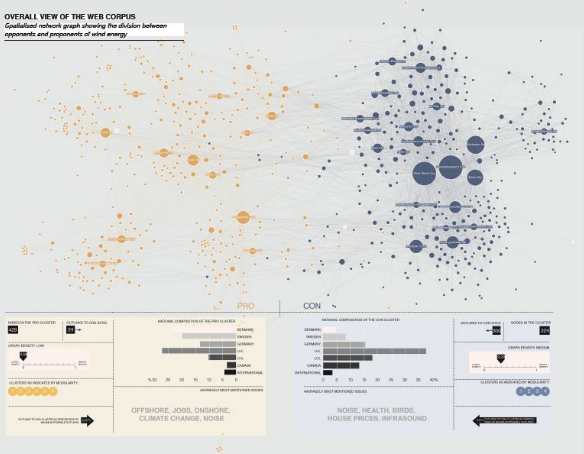
| | CANADA | DENMARK | GERMANY | UNITED STATES | SWEDEN | UNITED KINGDOM | INTERNATIONAL |
|---|---|---|---|---|---|---|---|
| Number of websites PRO wind energy | 11 | 104 | 54 | 43 | 85 | 115 | 17 |
| Number of websites CON wind energy | 42 | 15 | 50 | 58 | 27 | 122 | 10 |
| Number of undecided websites | | 1 | | 1 | | 2 | 1 |

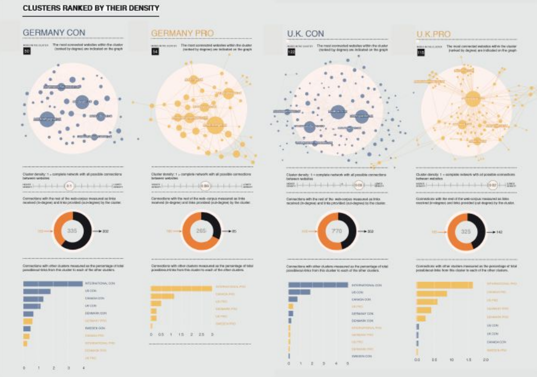
TOTAL
758
WEBSITES

### OVERALL VIEW OF THE WEB CORPUS

*Spatialised network graph showing the division between opponents and proponents of wind energy*




### CLUSTERS RANKED BY THEIR DENSITY




### DISCURSIVE REGIONS IN THE WEB CORPUS

*Nodes sized by the number of pages mentioning an issue term as a percentage of the total number of pages on the website*


### PROFILING INDIVIDUAL WEBSITES WITH THE ISSUE DICTIONARY

*Pages mentioning issue terms as a percentage of the total number of pages on a website*


*The kostervig case*


### CROSS COUNTING ISSUE TERMS IN THE WEB CORPUS

*How are different wind turbine projects associated with different issues?*


Wind2050 (www.wind2050.dk) is sponsored by the Danish Council for Strategic Research and led by the Danish Technical University
Principal investigator: Kristian Borch (krbo@dtu.dk)

**Author: Anders Kristian Munk (akm@learning.aau.dk)**
Data visualization and design: Anne-Kirstine Bøcher Ellern (www.akellern.com)
Student collaborators: Loretta Møller, Linn Wagner Korsgaard & Line Østereng Jørgensen
November 2014

The Danish Council for Strategic Research

DTU Management Engineering
Department of Management Engineering

AALBORG UNIVERSITY COPENHAGEN

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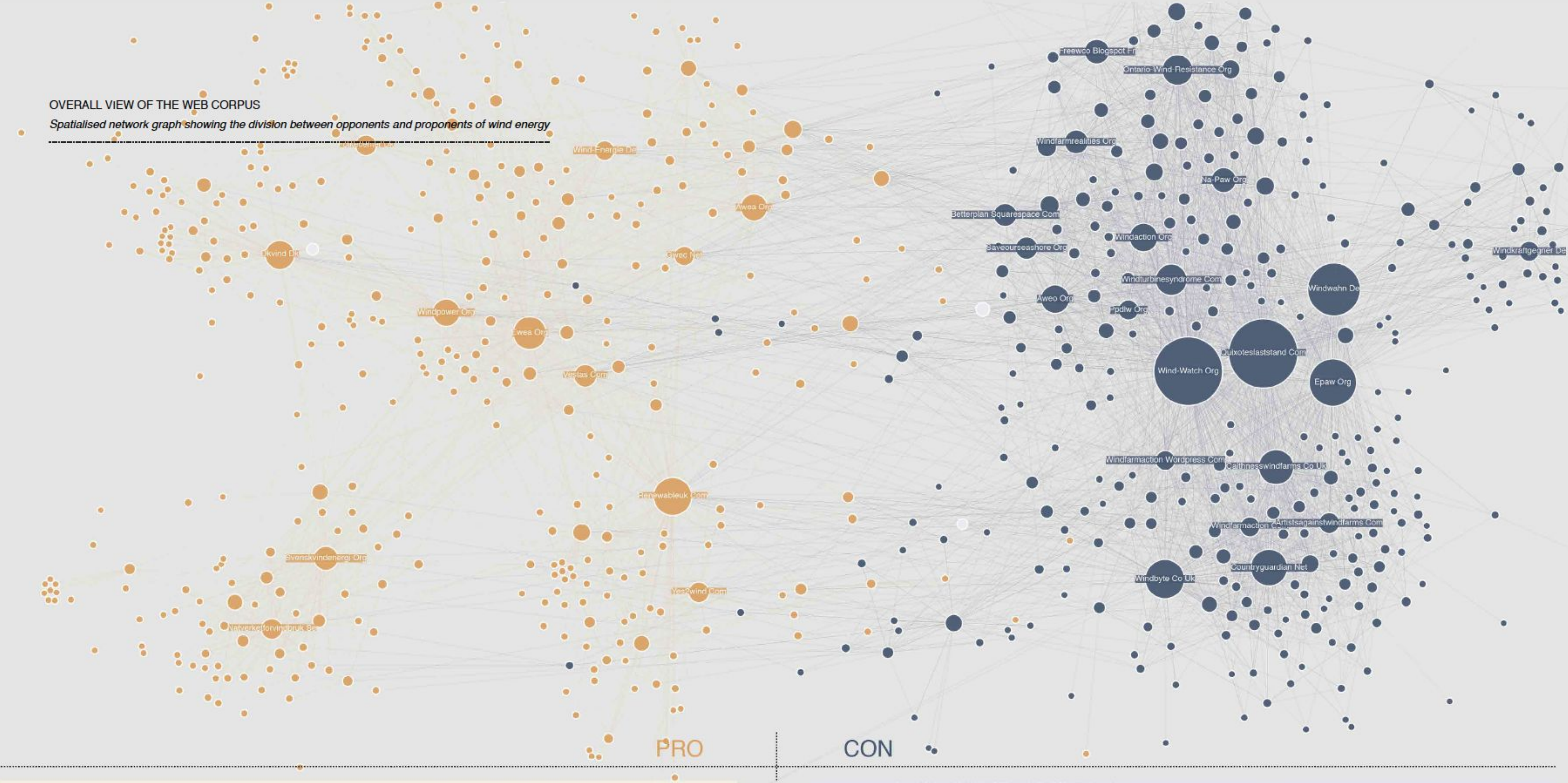



NODES IN THE PRO CLUSTER **429**

OUTLINKS TO CON SITES **24** →

GRAPH DENSITY. LOW

NATIONAL COMPOSITION OF THE PRO CLUSTER

DENMARK
SWEDEN
GERMANY

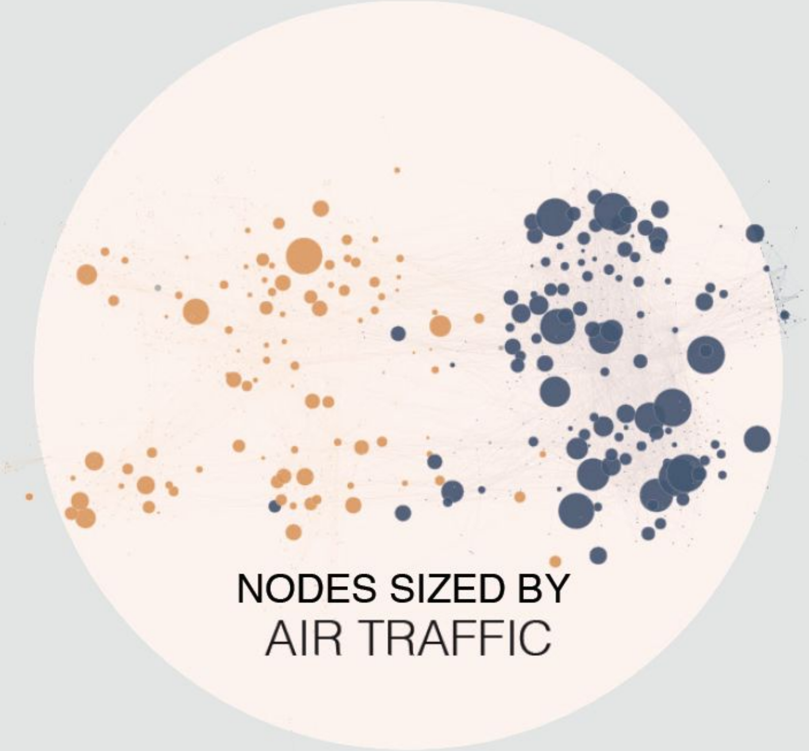
NATIONAL COMPOSITION OF THE CON CLUSTER

DENMARK
SWEDEN
GERMANY

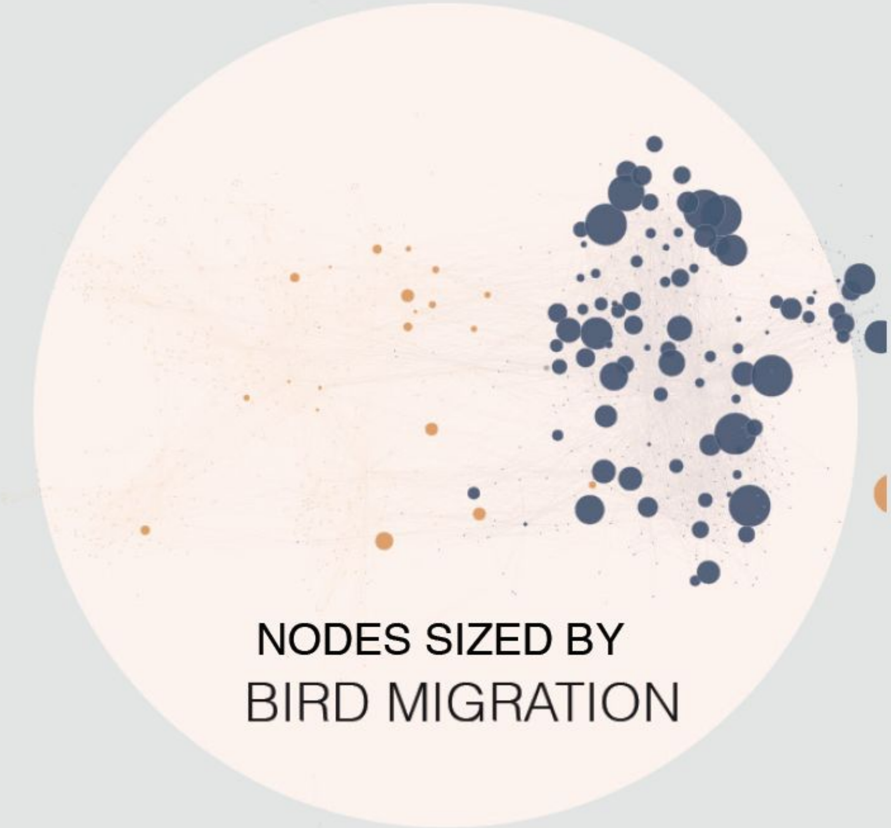
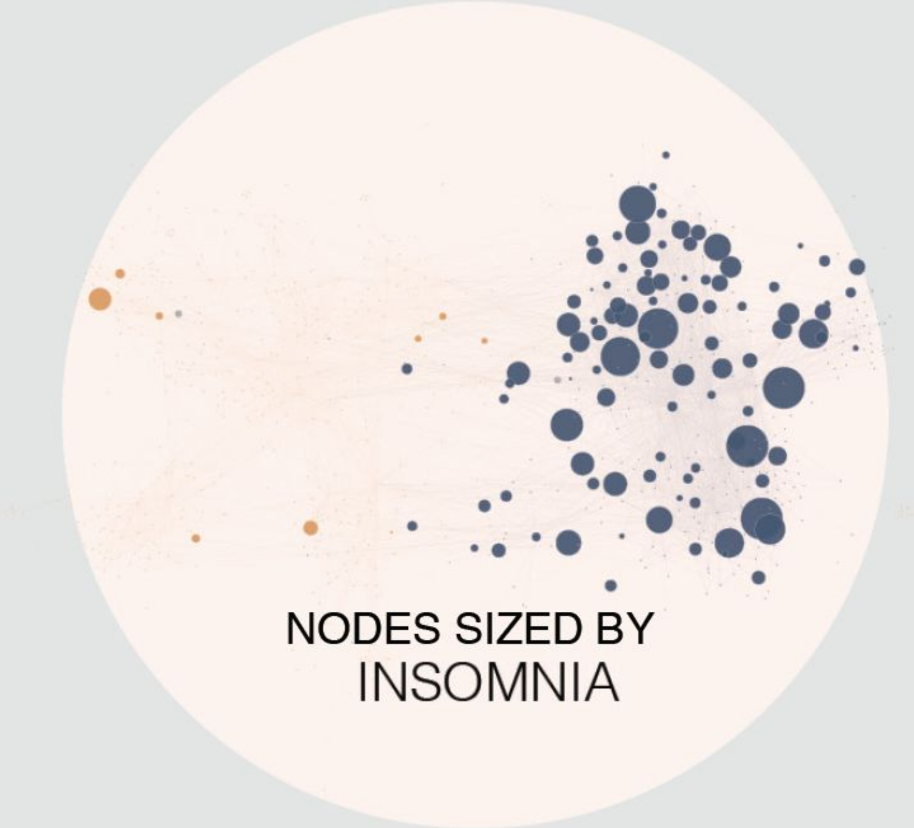
OUTLINKS TO CON SITES ← **300**

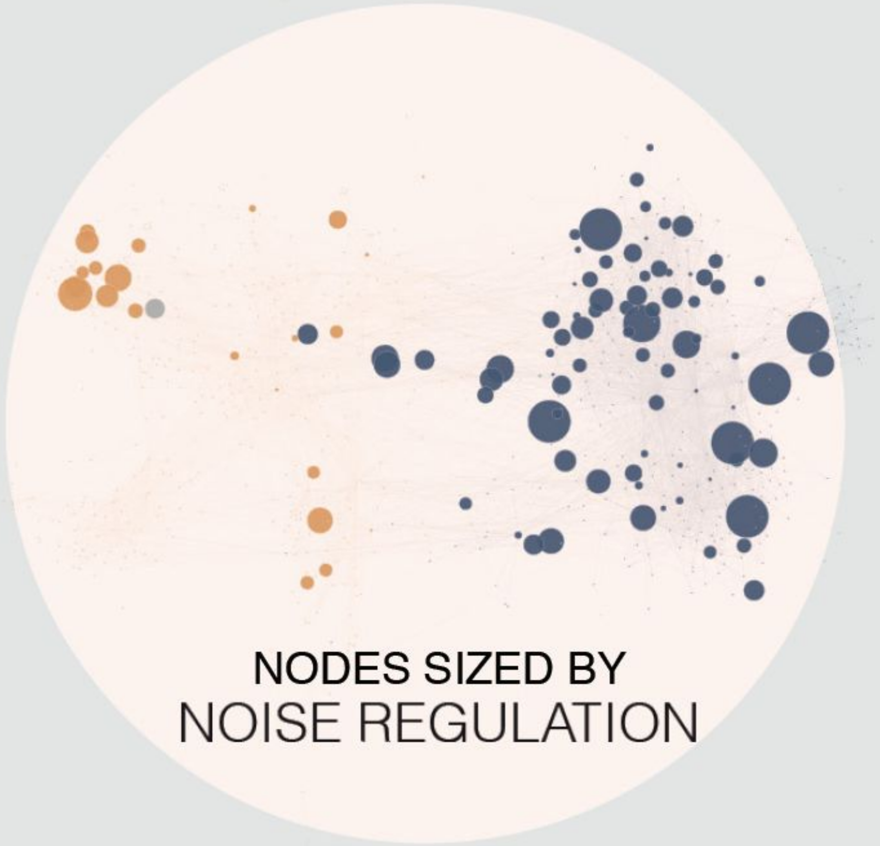
NODES IN THE CLUSTER **324**

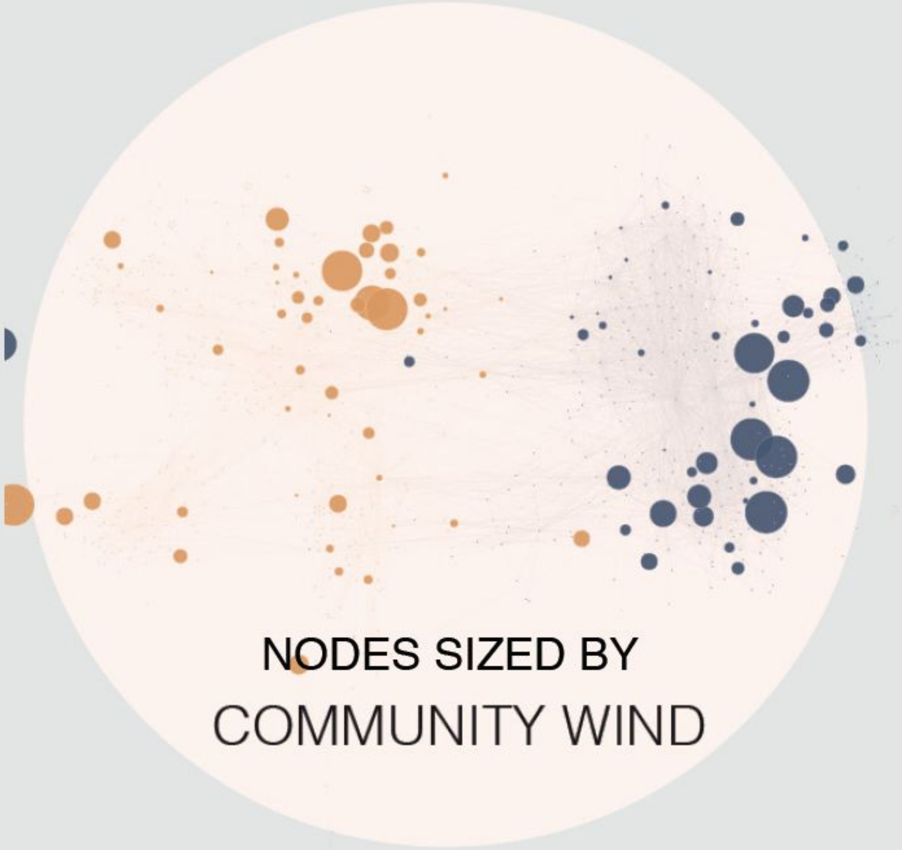
GRAPH DENSITY. MEDIUM


NODES SIZED BY
AIR TRAFFIC


NODES SIZED BY
SETBACK DISTANCE


NODES SIZED BY
BIRD MIGRATION

NODES SIZED BY
INSOMNIA


NODES SIZED BY
NOISE REGULATION

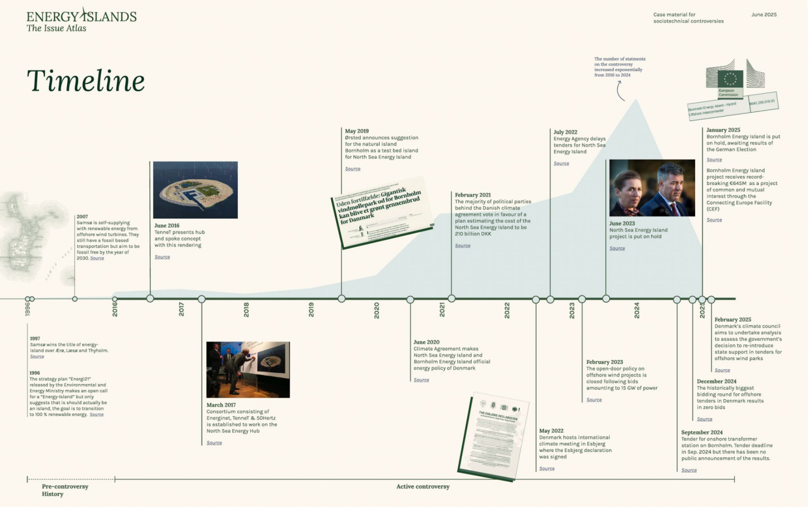

NODES SIZED BY
COMMUNITY WIND

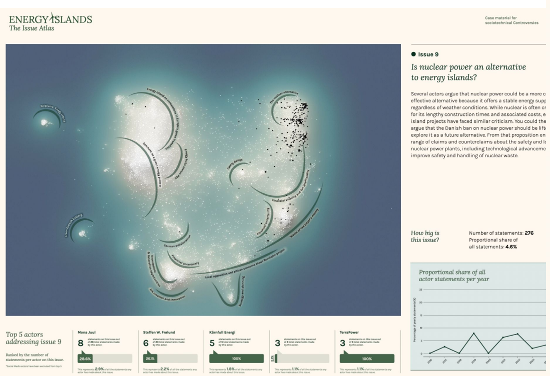
# From the policy brief exam assignment:

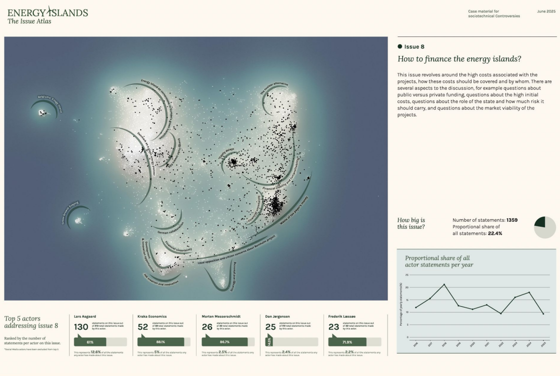
| | |
|---|---|
| **Methodology**<br><br>Goal: Explain how you have analysed the data in the issue atlas on the Danish energy islands and why. | Explain how you have searched the dataset from the issue atlas of the Danish Energy Islands to find the actor statements you will be analysing.<br><br>Justify why this search strategy addresses the focus of your policy brief (considering the nature of your proposal for improved sustainability and your chosen decision maker).<br><br>Present the data visualizations you will be using in the analysis and explain how they were made, how they should be read and how they helped you select actor statements.<br><br>Explain how your methodology follows principles for controversy mapping using methodological concepts from part 2 of the course. |

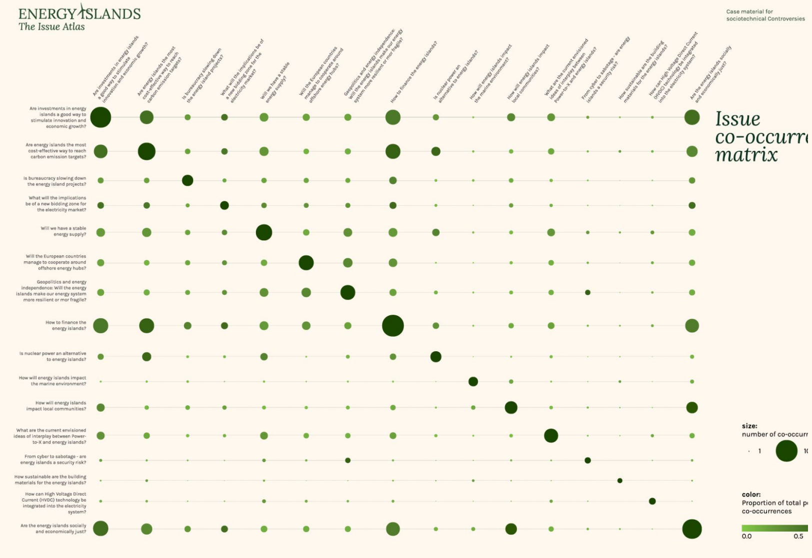
# The energy island issue atlas

This is where you will find information about how the actor statements were collected, how the issues were selected and tagged, how to read the maps, what kind of metadata you have available, etc.


ENERGY ISLANDS
The Issue Atlas
Case material for sociotechnical controversies
June 2025
Timeline
Pre-controversy History
Active controversy


ENERGY ISLANDS
The Issue Atlas
Issue 9
Is nuclear power an alternative to energy islands?
How big is this issue?
Proportional share of all actor statements per year
Top 5 actors addressing issue 9


ENERGY ISLANDS
The Issue Atlas
Issue 8
How to finance the energy islands?
How big is this issue?
Proportional share of all actor statements per year
Top 5 actors addressing issue 8


ENERGY ISLANDS
The Issue Atlas
Case material for sociotechnical Controversies
Issue co-occurr
matrix
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number of co-occurr
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color:
Proportion of total p
co-occurrences
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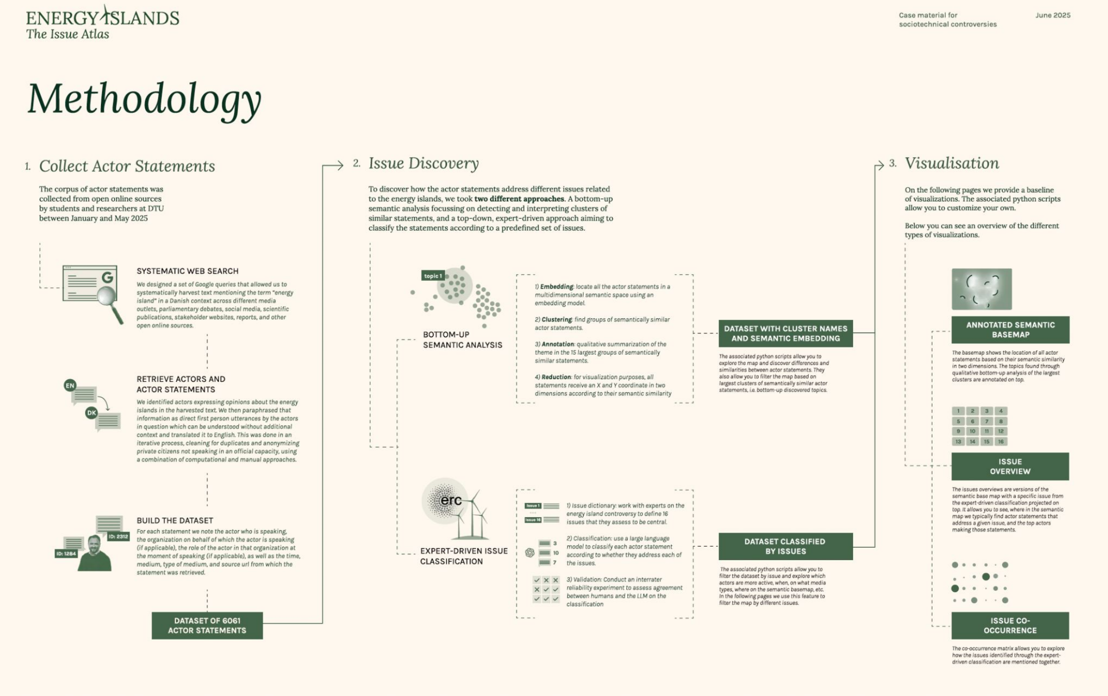
0.5


ENERGY ISLANDS
The Issue Atlas
Case material for sociotechnical controversies
June 2025
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1. Collect Actor Statements
The corpus of actor statements was collected from open online sources by students and researchers at DTU between January and May 2025
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We designed a set of Google queries that allowed us to systematically harvest text mentioning the term "energy island" in a Danish context across different media outlets, parliamentary debates, social media, scientific publications, stakeholder websites, reports, and other open online sources.
RETRIEVE ACTORS AND ACTOR STATEMENTS
EN
DK
We identified actors expressing opinions about the energy islands in the harvested text. We then paraphrased that information as direct first person utterances by the actors in question which can be understood without additional context and translated it to English. This was done in an iterative process, cleaning for duplicates and anonymizing private citizens not speaking in an official capacity, using a combination of computational and manual approaches.
BUILD THE DATASET
ID: 2312
ID: 1284
For each statement we note the actor who is speaking, the organization on behalf of which the actor is speaking (if applicable), the role of the actor in that organization at the moment of speaking (if applicable), as well as the time, medium, type of medium, and source url from which the statement was retrieved.
DATASET OF 6061 ACTOR STATEMENTS
2. Issue Discovery
To discover how the actor statements address different issues related to the energy islands, we took two different approaches. A bottom-up semantic analysis focussing on detecting and interpreting clusters of similar statements, and a top-down, expert-driven approach aiming to classify the statements according to a predefined set of issues.
topic 1
BOTTOM-UP SEMANTIC ANALYSIS
1) Embedding: locate all the actor statements in a multidimensional semantic space using an embedding model.
2) Clustering: find groups of semantically similar actor statements.
3) Annotation: qualitative summarization of the theme in the 15 largest groups of semantically similar statements.
4) Reduction: for visualization purposes, all statements receive an X and Y coordinate in two dimensions according to their semantic similarity
DATASET WITH CLUSTER NAMES AND SEMANTIC EMBEDDING
The associated python scripts allow you to explore the map and discover differences and similarities between actor statements. They also allow you to filter the map based on largest clusters of semantically similar actor statements, i.e. bottom-up discovered topics.
erc
EXPERT-DRIVEN ISSUE CLASSIFICATION
Issue 1
Issue 16
1) Issue dictionary: work with experts on the energy island controversy to define 16 issues that they assess to be central.
3
10
7
2) Classification: use a large language model to classify each actor statement according to whether they address each of the issues.
3) Validation: Conduct an interrater reliability experiment to assess agreement between humans and the LLM on the classification
DATASET CLASSIFIED BY ISSUES
The associated python scripts allow you to filter the dataset by issue and explore which actors are more active, when, on what media types, where on the semantic basemap, etc. In the following pages we use this feature to filter the map by different issues.
3. Visualisation
On the following pages we provide a baseline of visualizations. The associated python scripts allow you to customize your own.
Below you can see an overview of the different types of visualizations.
ANNOTATED SEMANTIC BASEMAP
The basemap shows the location of all actor statements based on their semantic similarity in two dimensions. The topics found through qualitative bottom-up analysis of the largest clusters are annotated on top.
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
ISSUE OVERVIEW
The issues overviews are versions of the semantic base map with a specific issue from the expert-driven classification projected on top. It allows you to see, where in the semantic map we typically find actor statements that address a given issue, and the top actors making those statements.
ISSUE CO-OCCURRENCE
The co-occurrence matrix allows you to explore how the issues identified through the expert-driven classification are mentioned together.

# Timeline

The number of statments on the controversy increased exponentially from 2016 to 2024

**1996**
The strategy plan "Energi21" released by the Environmental and Energy Ministry makes an open call for a "Energy-Island" but only suggests that is should actually be an island, the goal is to transition to 100 % renewable energy. Source

**1997**
Samsø wins the title of energy-island over Ærø, Læsø and Thyholm.
Source

**2007**
Samsø is self-supplying with renewable energy from offshore wind turbines. They still have a fossil based transportation but aim to be fossil free by the year of 2030. Source

**June 2016**
TenneT presents hub and spoke concept with this rendering
Source

**March 2017**
Consortium consisting of Energinet, TenneT & 50Hertz is established to work on the North Sea Energy Hub
Source

**May 2019**
Ørsted announces suggestion for the natural island Bornholm as a test bed island for North Sea Energy Island
Source


**June 2020**
Climate Agreement makes North Sea Energy Island and Bornholm Energy Island official energy policy of Denmark
Source

**February 2021**
The majority of political parties behind the Danish climate agreement vote in favour of a plan estimating the cost of the North Sea Energy Island to be 210 billion DKK
Source


**May 2022**
Denmark hosts international climate meeting in Esbjerg where the Esbjerg declaration was signed
Source

**July 2022**
Energy Agency delays tenders for North Sea Energy Island
Source

**February 2023**
The open-door policy on offshore wind projects is closed following bids amounting to 15 GW of power
Source

**June 2023**
North Sea Energy Island project is put on hold
Source

**September 2024**
Tender for onshore transformer station on Bornholm. Tender deadline in Sep. 2024 but there has been no public announcement of the results.
Source

**December 2024**
The historically biggest bidding round for offshore tenders in Denmark results in zero bids
Source


**January 2025**
Bornholm Energy Island is put on hold, awaiting results of the German Election
Source

Bornholm Energy Island project receives record-breaking €645M as a project of common and mutual interest through the Connecting Europe Facility (CEF)
Source

**February 2025**
Denmark's climate council aims to undertake analysis to assess the government's decision to re-introduce state support in tenders for offshore wind parks
Source

1996 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025

**Pre-controversy History**

**Active controversy**

# Methodology

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### BOTTOM-UP SEMANTIC ANALYSIS

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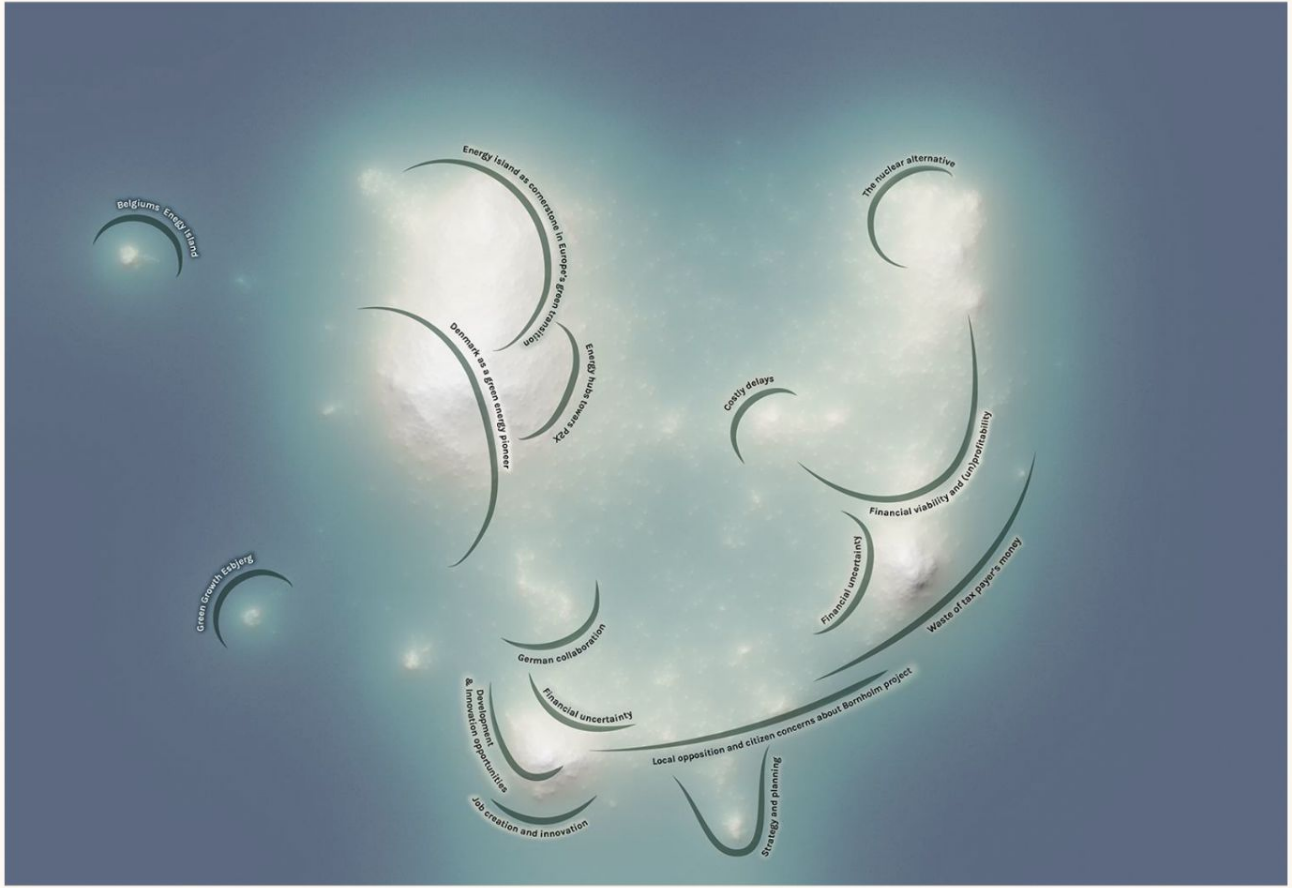
**ISSUE CO-OCCURRENCE**

*The co-occurrence matrix allows you to explore how the issues identified through the expert-driven classification are mentioned together.*

# Annotated Basemap




The statements are more or less grouped in clusters in the map, based on their semantic similarity. These clusters have an approximate centerpoint - the centroid. Around each centroid there are between 60 and 200 very closely related statements.

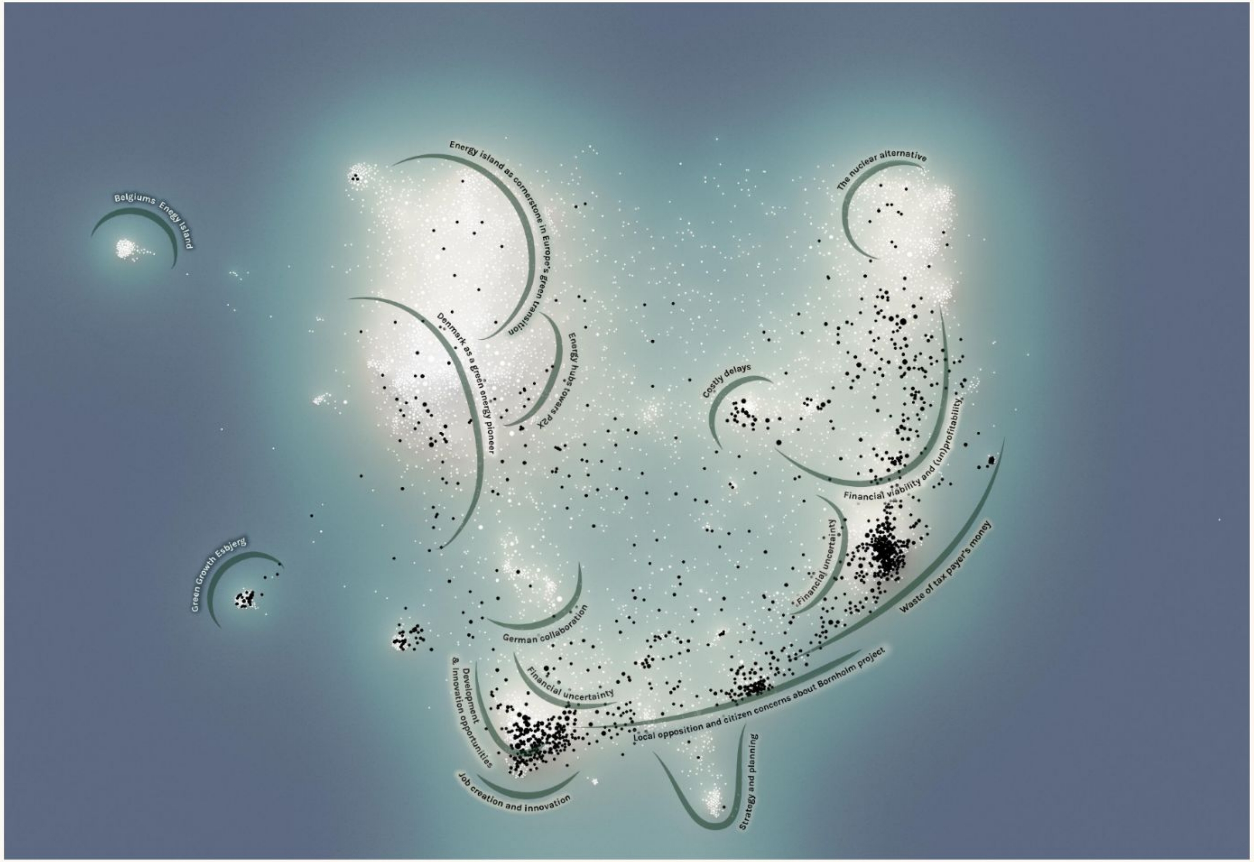
We use an LLM to group them together in annotation batches and prompt the LLM to summarize the content of each cluster in maximum 6 words. These short annotations are the ones you see in the map.

The vague dynamics around them are neither closed areas nor near exact shapes - they are estimates of the radius around the centroid of the cluster where related statements are located.

**Issue 16**

# *Are the energy islands socially and economically just?*

Who will gain and who will bear the burdens of realizing the energy islands? These discussions revolve around the distribution of benefits, financial burdens, and the overall impact on society. Will the energy islands generate a net socio-economic benefit? To what extent should taxpayers bear the burden? Will the projects create jobs and for whom? Will the local communities be properly involved? Is there some form of co-ownership and how can you ensure social acceptance?




● **Issue 16**

## *Are the energy islands socially and economically just?*

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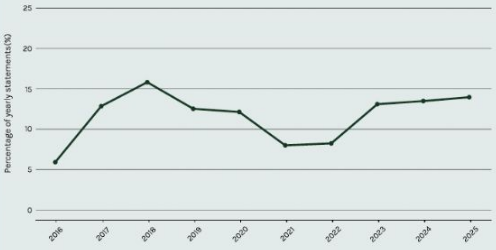
*How big is this issue?*

Number of statements: **1140**
Proportional share of all statements: **12.1%**

*Proportional share of all actor statements per year*




*Top 5 actors addressing issue 16*

Ranked by the number of statements per actor on this issue.

*Social Media actors have been excluded from top 5

**Kraka Economics**

50 statements on this issue out of **59** total statements made by this actor.

84.7%

This represents **4.4%** of all the statements any actor has made about this issue.

**Lars Aagaard**

50 statements on this issue out of **213** total statements made by this actor.

23.5%

This represents **4.4%** of all the statements any actor has made about this issue.

**Jacob Trøst**

44 statements on this issue out of **82** total statements made by this actor.

53.7%

This represents **3.9%** of all the statements any actor has made about this issue.

**Søren Møller Christensen**

23 statements on this issue out of **60** total statements made by this actor.

38.3%

This represents **2%** of all the statements any actor has made about this issue.

**Morten Messerschmidt**

22 statements on this issue out of **30** total statements made by this actor.

73.3%

This represents **1.9%** of all the statements any actor has made about this issue.

*"I insist on prioritizing local business development on Bornholm well before the Energy Island project fully arrives. Bornholm-based companies must be engaged in the tasks associated with such a large-scale construction so that as many assignments as possible remain with local enterprises."*

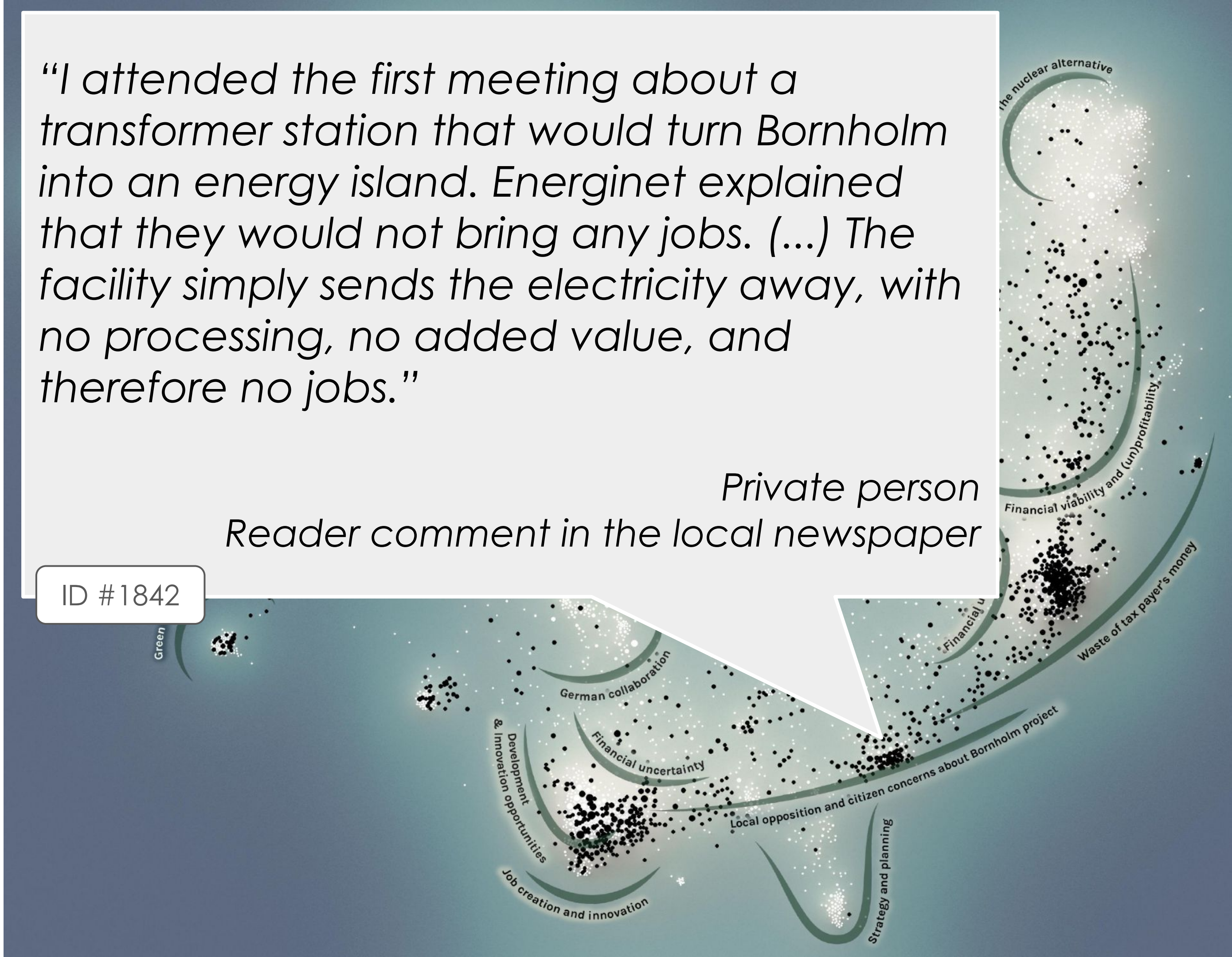
*Linda Kofoed Persson*
*Conservative local politician*
*Municipality of Bornholm*

ID #7030

*"I attended the first meeting about a transformer station that would turn Bornholm into an energy island. Energinet explained that they would not bring any jobs. (...) The facility simply sends the electricity away, with no processing, no added value, and therefore no jobs."*

*Private person*
*Reader comment in the local newspaper*

ID #1842

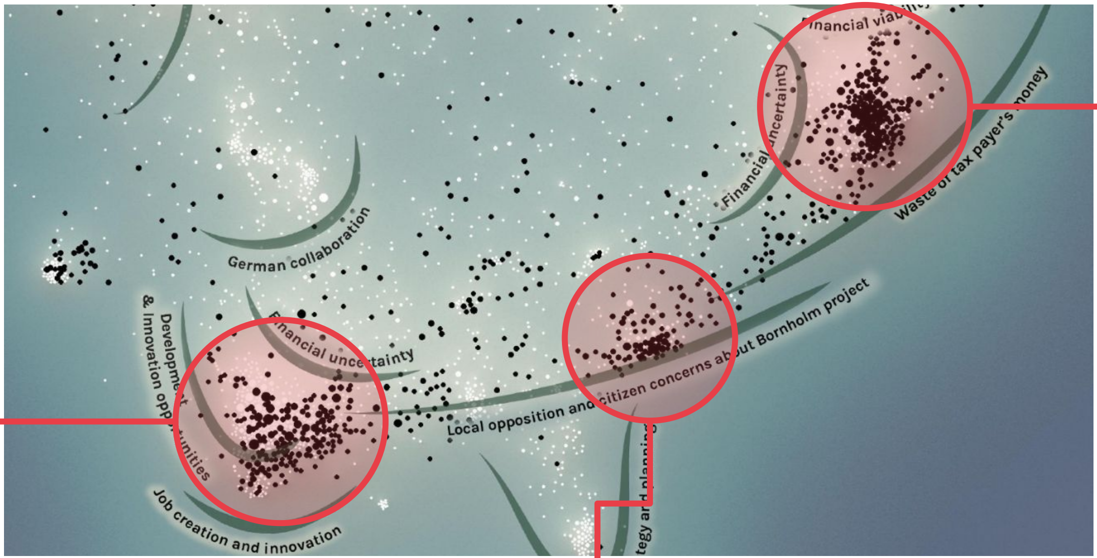



*"The energy islands in Denmark are an economic disaster unfolding slowly. Last week, Denmark's two energy island projects, North Sea and Bornholm, were once again deemed uneconomical. Now, Germany's interest in these projects only serves to confirm that they are a disaster."*

*Johan Christian Sollid*
*Founder and chairman of Atomkraft Ja Tak (Nuclear Power Yes Please)*

ID #873

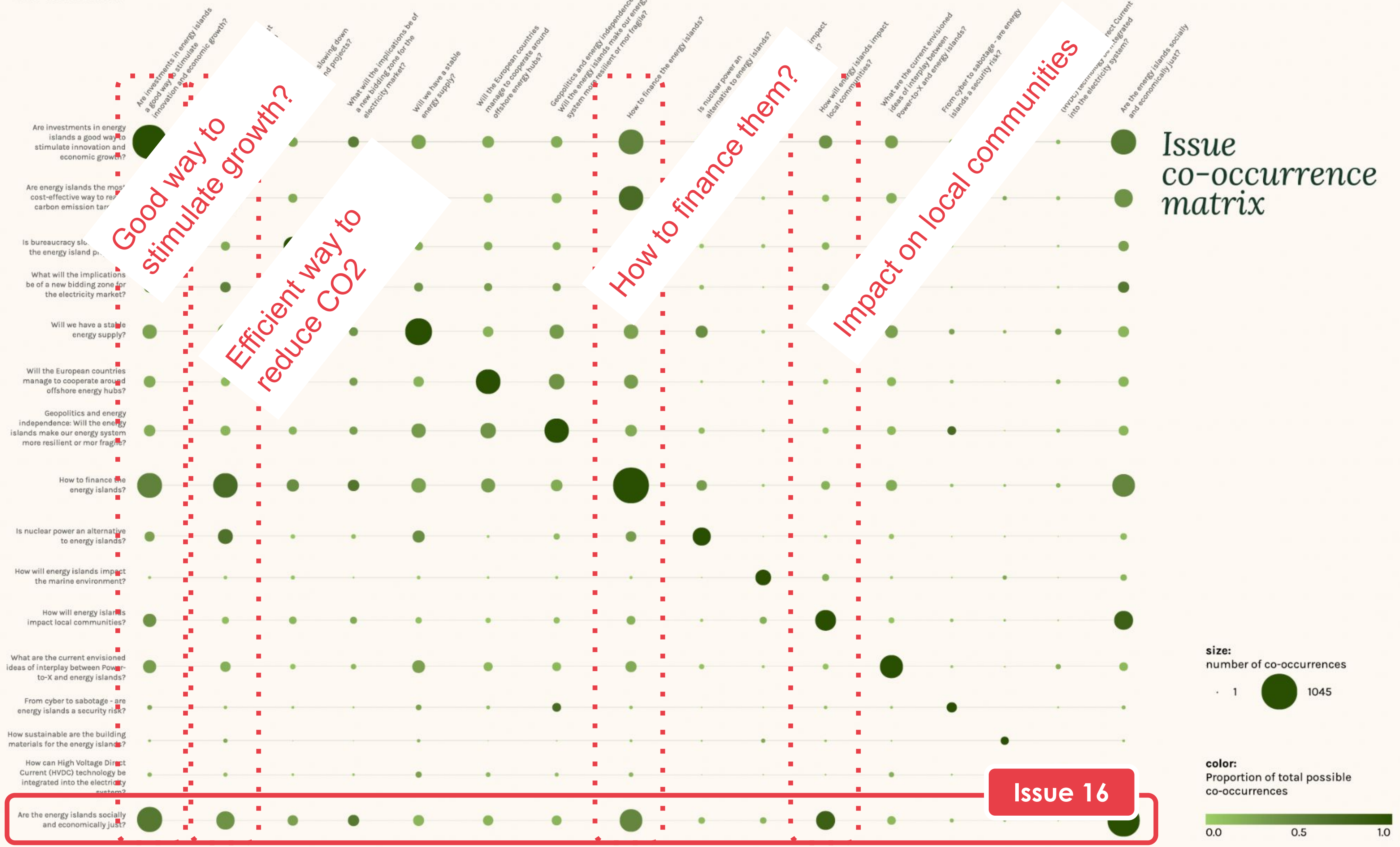
**Hypothesis:** There are three versions of the issue: Are the energy islands socially and economically just? (Issue 16)




**Version 1:** They are good for local businesses. They will generate local jobs and innovation.

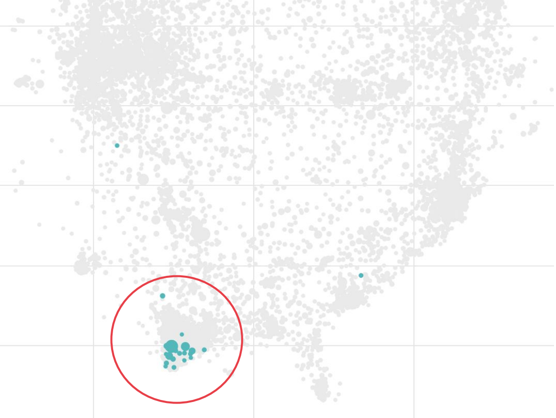
**Version 2:** The local community will not benefit. The jobs will be created elsewhere.

**Version 3:** It is unfair to the taxpayers when the experts (Kraka) have told us it will be a bad investment.


Issue co-occurrence matrix
Are investments in energy islands a good way to stimulate innovation and economic growth?
Are energy islands the most cost-effective way to reach carbon emission targets?
Is bureaucracy slowing down the energy island projects?
What will the implications be of a new bidding zone for the electricity market?
Will we have a stable energy supply?
Will the European countries manage to cooperate around offshore energy hubs?
Geopolitics and energy independence: Will the energy islands make our energy system more resilient or mor fragile?
How to finance the energy islands?
Is nuclear power an alternative to energy islands?
How will energy islands impact the marine environment?
How will energy islands impact local communities?
What are the current envisioned ideas of interplay between Power-to-X and energy islands?
From cyber to sabotage - are energy islands a security risk?
How sustainable are the building materials for the energy islands?
How can High Voltage Direct Current (HVDC) technology be integrated into the electricity system?
Are the energy islands socially and economically just?
Good way to stimulate growth?
Efficient way to reduce CO2
How to finance them?
Impact on local communities
Issue 16
size:
number of co-occurrences
1
1045
color:
Proportion of total possible co-occurrences
0.0
0.5
1.0

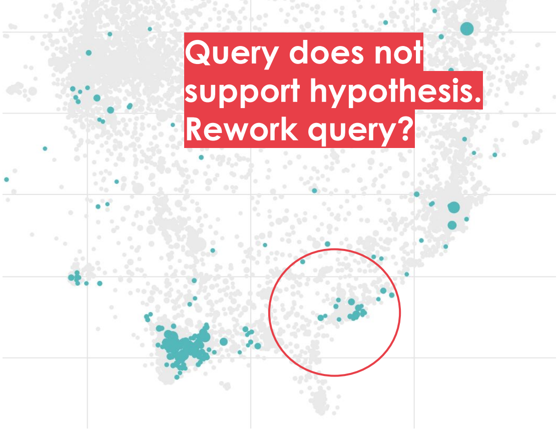
**Hypothesis:** There are three versions of the issue: Are the energy islands socially and economically just? (Issue 16)

Statements labelled **Issue 16** AND containing the phrase "**local business**"



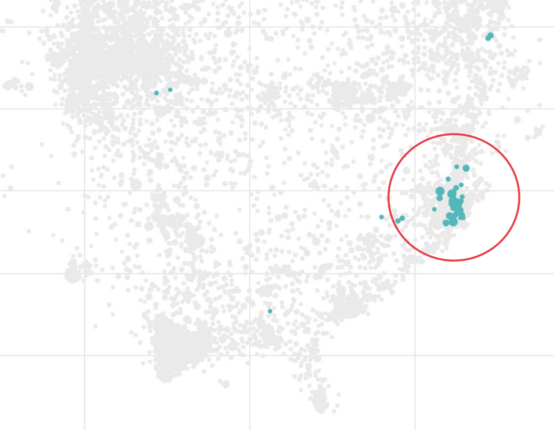
**Version 1:** They are good for local businesses. They will generate local jobs and innovation.

Statements labelled **Issue 16** AND containing the phrase "**jobs**"



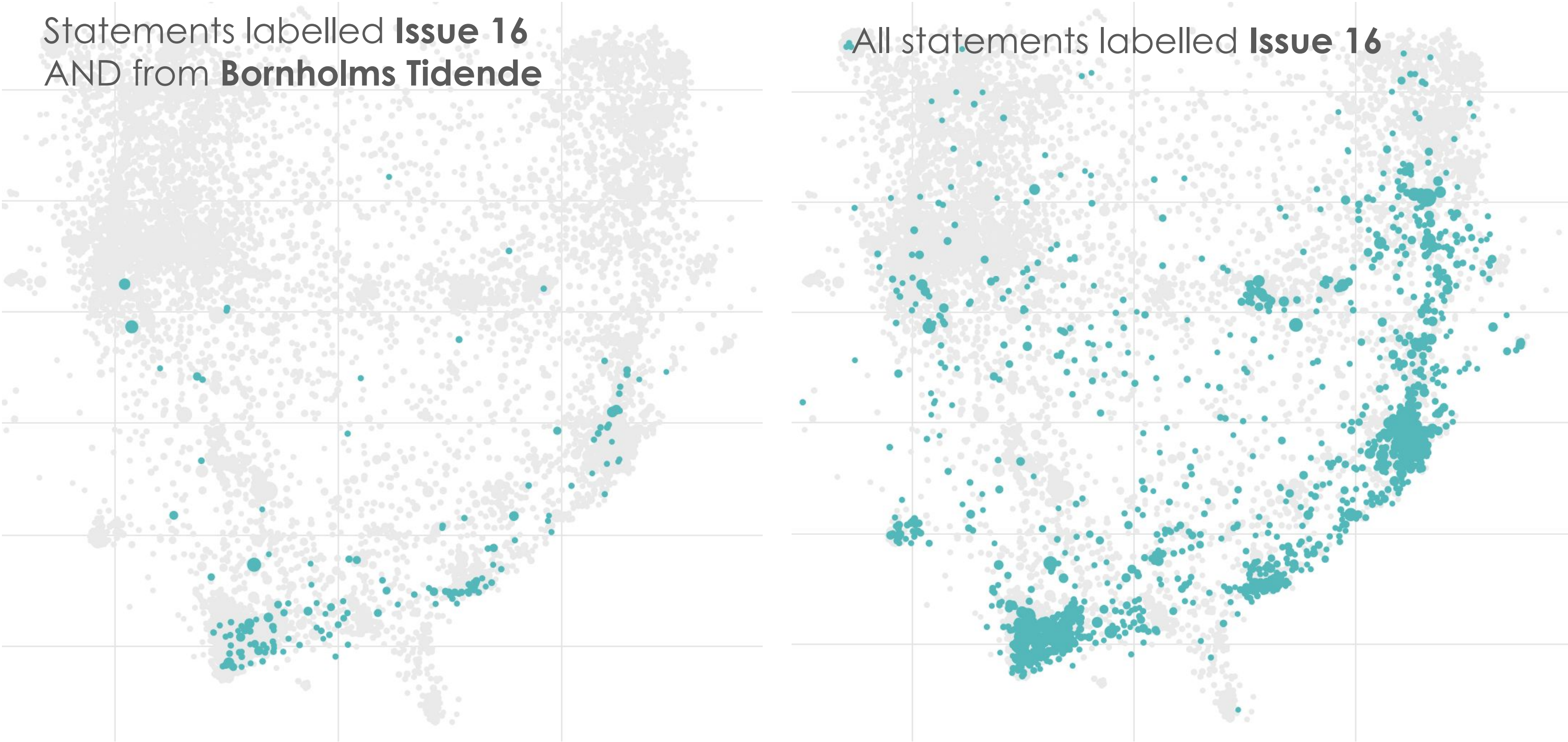

**Version 2:** The local community will not benefit. The jobs will be created elsewhere.

Statements labelled **Issue 16** AND containing the phrase "**Kraka**"

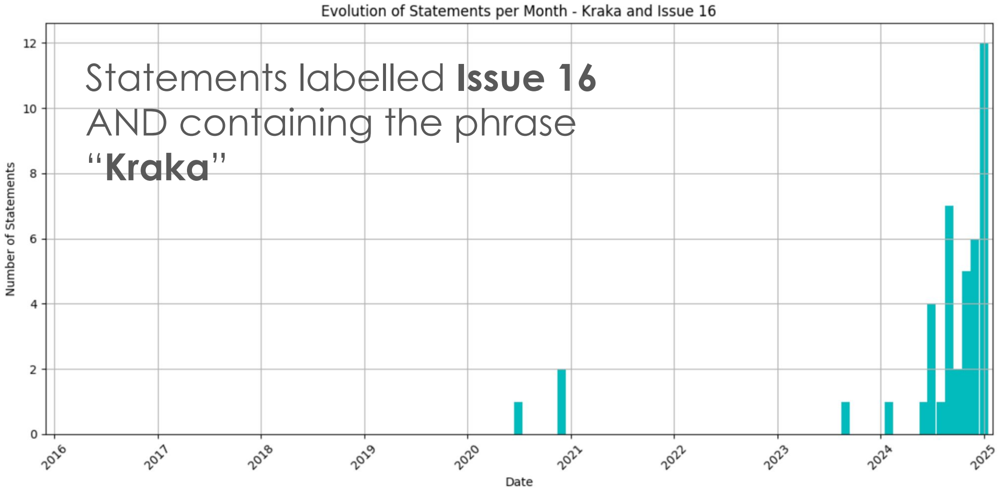


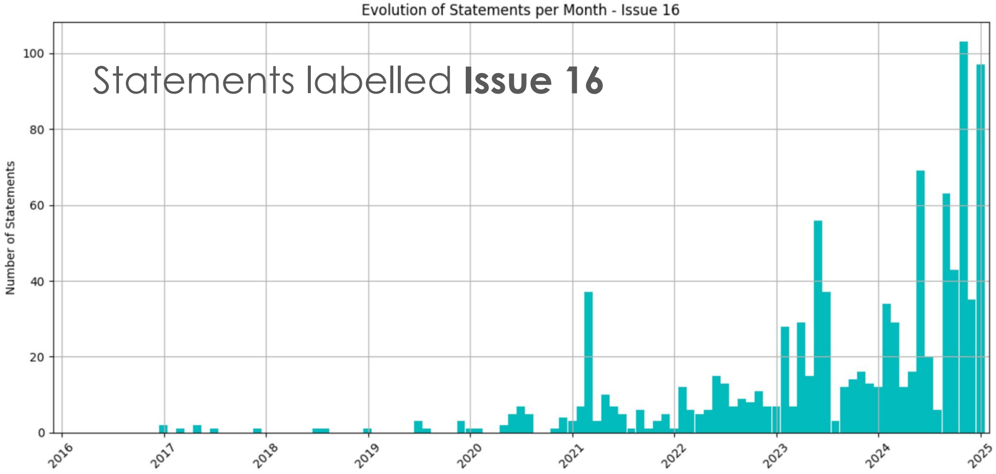
**Version 3:** It is unfair to the taxpayers when the experts (Kraka) have told us it will be a bad investment.

**Hypothesis:** Versions 1 and 2 of Issue 16 are more locally driven.



**Hypothesis:** Version 3 of Issue 16 emerged later and has been particularly fuelled by the publication of the critical Kraka report on the economy of the energy islands in 2024.



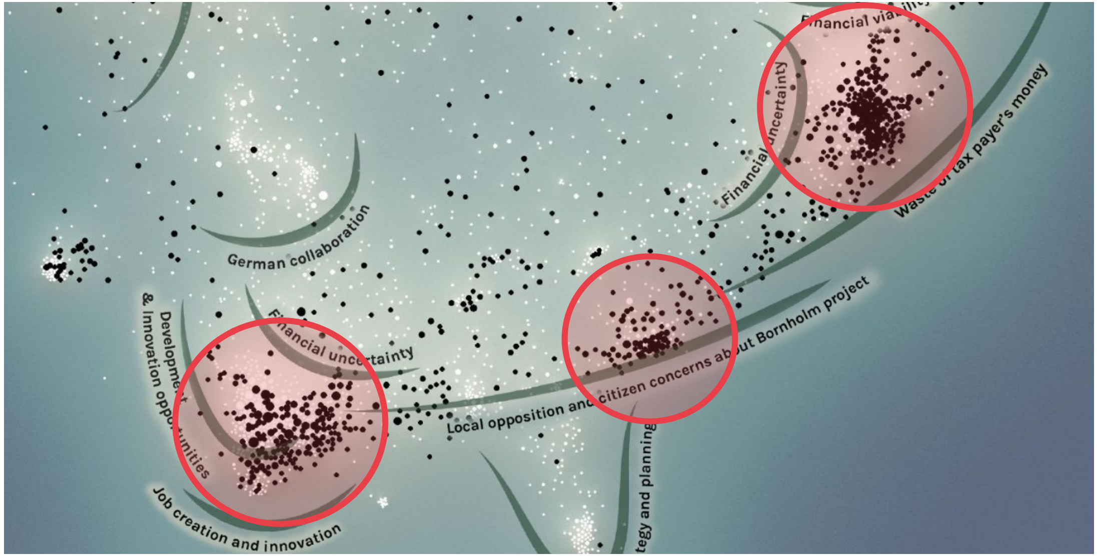



Your queries and visualizations are supposed to help you select actor statements that you can then read in your analysis.

# Examples of possible questions for the analysis:

- What are the underlying ethical frameworks expressed in these different versions of the energy islands as socially and economically just?
- Do they align with different different perceptions of risk?
- What are the implications of these different problem framings for problem wickedness?
- How do they discuss expertise and who do they consider to be relevant experts?

**NB:** Whether these are good questions to pursue or not depends on what you find. Iteration will be necessary.



Your query strategy must be tailored to the focus of your policy brief (not just looking for a random issue like above)